

1 Solution — GIAM 1.4

Problem 11

1.1 Problem

True or false?

$$\left\lfloor \frac{n}{d} \right\rfloor < \left\lceil \frac{n}{d} \right\rceil \quad \text{for all integers } n \text{ and } d > 0.$$

Support your claim.

1.2 Verdict: False

The *strict* inequality fails whenever d divides n evenly. In that case n/d is itself an integer, and the floor and ceiling of an integer are *equal* — not strictly ordered.

1.3 A Counterexample

Take $n = 6$, $d = 3$. Then $n/d = 6/3 = 2$, an integer, so

$$\left\lfloor \frac{6}{3} \right\rfloor = \lfloor 2 \rfloor = 2, \quad \left\lceil \frac{6}{3} \right\rceil = \lceil 2 \rceil = 2.$$

The claim asserts $2 < 2$, which is **false**. So the universal statement is disproved by this single counterexample. (Any multiple of d works: $n = 0, d = 1$ gives $0 < 0$; $n = d$ gives $1 < 1$; and so on.)

1.4 The Correct Statement

What *is* always true is the **non-strict** inequality, together with a precise description of when it is strict:

$$\left\lfloor \frac{n}{d} \right\rfloor \leq \left\lceil \frac{n}{d} \right\rceil, \quad \text{with} \quad \begin{cases} \left\lfloor \frac{n}{d} \right\rfloor = \left\lceil \frac{n}{d} \right\rceil & \text{if } d \mid n, \\ \left\lfloor \frac{n}{d} \right\rfloor < \left\lceil \frac{n}{d} \right\rceil & \text{if } d \nmid n. \end{cases}$$

The reason is the basic floor/ceiling dichotomy: for any real x ,

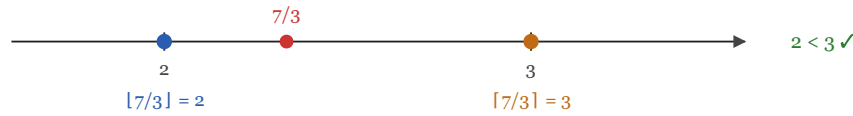
$$\lceil x \rceil - \lfloor x \rfloor = \begin{cases} 0 & \text{if } x \in \mathbb{Z}, \\ 1 & \text{if } x \notin \mathbb{Z}. \end{cases}$$

Here $x = n/d$ is an integer exactly when $d \mid n$. So the strict inequality $\lfloor n/d \rfloor < \lceil n/d \rceil$ holds **if and only if** $d \nmid n$ — *not* for all n, d .

1.5 The Picture

The gap $\lceil x \rceil - \lfloor x \rfloor$ is 1 when x is not an integer, but 0 when it is — so strict $<$ can fail.

When $d \nmid n$ (e.g. $7/3 \approx 2.33$): $\lfloor \cdot \rfloor < \lceil \cdot \rceil$ holds.



When $d \mid n$ (e.g. $6/3 = 2$): $\lfloor \cdot \rfloor = \lceil \cdot \rceil$ — strict $<$ FAILS.



So the claim is *FALSE*. The true statement is $\lfloor n/d \rfloor \leq \lceil n/d \rceil$, with equality exactly when $d \mid n$.

Figure 1.1: Top: for $7/3 \approx 2.33$ (not an integer), the floor 2 and ceiling 3 differ by 1, so $\lfloor \cdot \rfloor < \lceil \cdot \rceil$ holds. Bottom: for $6/3 = 2$ (an integer), the floor and ceiling coincide at 2, so $\lfloor \cdot \rfloor < \lceil \cdot \rceil$ fails — the counterexample. The strict inequality holds only when $d \nmid n$.

1.6 Remark — The Gap Is Always 0 or 1

The whole question turns on a single clean fact: a real number is squeezed between its floor and ceiling,

$$\lfloor x \rfloor \leq x \leq \lceil x \rceil,$$

and these two integer bounds are *never more than 1 apart*. They are equal when x already sits on an integer (nothing to round), and exactly 1 apart otherwise (rounding down vs. rounding up land on adjacent integers). The proposed inequality mistakes " $\leq$ " for " $<$ " by forgetting the equal case — a textbook off-by-the-boundary slip.

1.7 Remark — It’s the Remainder Again

The condition $d \mid n$ is just “the remainder is zero,” tying this back to division-with-remainder (and to Problem 10). Write $n = qd + r$ with $0 \leq r < d$:

- $r = 0$ ($d \mid n$): $n/d = q$ exactly, so $\lfloor n/d \rfloor = \lceil n/d \rceil = q$. Equal.
- $r > 0$ ($d \nmid n$): $n/d = q + r/d$ with $0 < r/d < 1$, so $\lfloor n/d \rfloor = q$ and $\lceil n/d \rceil = q + 1$. Strictly less.

This is the *same* dichotomy that decided whether Problem 10 needed an “extra bag”: a nonzero remainder is exactly what makes ceiling exceed floor.

1.8 Remark — How to Fix the Claim

Two repaired statements are both true, and each is useful:

- **Weaken the relation:** $\lfloor n/d \rfloor \leq \lceil n/d \rceil$ for all integers n and $d > 0$. (Always holds.)
- **Restrict the hypothesis:** $\lfloor n/d \rfloor < \lceil n/d \rceil$ for all integers n and $d > 0$ with $d \nmid n$. (The strict version, correctly qualified.)

This mirrors the repair strategy from the quantifier sections (1.3) and the mercury problem: when a universal claim is refuted by boundary cases, you either *loosen the conclusion* ($<\rightarrow\leq$) or *shrink the domain* (exclude $d \mid n$). The original error is exactly the kind of overstatement careful notation is meant to catch.

1.9 Remark — Negative and Zero Cases Behave the Same

The counterexamples don’t depend on positivity. The claim allows *all* integers n (with only $d > 0$), and the equal-case appears throughout:

- $n = 0$: $0/d = 0$, so $\lfloor 0 \rfloor = \lceil 0 \rceil = 0$ for *every* d — infinitely many counterexamples at once.
- $n = -6, d = 3$: $-6/3 = -2$, and $\lfloor -2 \rfloor = \lceil -2 \rceil = -2$.

(Worth noting: floor and ceiling of *negative* non-integers still differ by 1 — e.g. $\lfloor -7/3 \rfloor = \lfloor -2.33 \rfloor = -3$ while $\lceil -7/3 \rceil = -2$, and $-3 < -2$ ✓. The sign of n never matters; only whether d divides n .) So the statement fails on a large, easily described family — every multiple of d — confirming it is false in the strongest way.